

Date: 09/06/2018 E.C

To: Final Year Project Committee

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Subject: Request for Approval to Continue Previous Project Title

Dear Respected Final Year Project Committee Members:

We respectfully submit this letter to request approval to continue our previously registered final year project entitled: "Road Traffic Control System."

The project was evaluated in the previous academic session, and based on the comments and professional recommendations given by our examiners, Advisor and stakeholders of the project we have improved the scope of the system by incorporating additional functional features and system refinements.

Accordingly, the updated project includes the following enhancements:

1. Integrate payment system
2. Complaint management system
3. Notification system
4. Hosting the service to Amhara Region Police Head Quarter

We appreciate your consideration and look forward to your approval.

Sincerely,

Name: Gebeyaw Belayneh ID No: 1410563 Signature: 

Name: Abreham Shiferaw ID No: 1404228 Signature: 

Name: Markos Teshome ID No: 1403700 Signature: 

Name: Bekalu Eshetie ID No: 1404579 Signature: 

Examiners Recommendation:

Examiner 1 Name: Tadele Mengstie Signature: 

Examiner 1 Name: Wondim Dessye Signature:

Project Proposal

Group Name: GAMB

Department: Software Engineering

Group Members: Gebeyaw | Abraham | Bekalu | Markos

Submission Date: 09/06/2018 E.C.

Proposed Project Titles:

1: Road Traffic Control System (RTCS)

Introduction

Transportation is very important for economic and social development. In the Amhara Region, the number of vehicles has increased rapidly due to population growth, urbanization, and expanding economic activities. Every day, many people use private cars, minibuses, trucks, motorcycles, Bajajs, and taxis for work, trade, education, and other services.

However, traffic management systems in the region still depend mainly on manual and paper-based processes. Traffic officers check driver licenses and vehicle documents visually and issue handwritten tickets. There is no centralized digital system that connects driver information, vehicle registration, insurance records, and violation history. This creates inefficiency, delays, and weak enforcement.

To address these issues, this project proposes the development of a **Road Traffic Control System (RTCS)** — a web and mobile-based system that digitizes traffic management and improves road safety in the Amhara Region.

Statement of the Problem

The current traffic enforcement system faces several challenges:

- There is no unified, real-time database connecting driver, vehicle, and violation records.
- Officers cannot instantly verify whether a license is genuine, suspended, or expired.
- Fake documents are difficult to detect due to the absence of digital verification tools.
- Manual traffic stops are slow and may cause congestion and delays.
- Paper-based tickets are easily lost, and unpaid fines are not properly tracked.
- There is limited transparency and accountability because traffic stops are not digitally recorded.
- Lack of reliable traffic data makes planning and policy decisions difficult.

Objectives

General Objective

To design and develop a web and mobile-based Road Traffic Control System (RTCS) that modernizes traffic law enforcement, ensures real-time data access, and improves accountability in the Amhara Region.

Specific Objectives

- Develop a centralized digital database for driver, vehicle, insurance, and violation information.
- Enable real-time verification of driver licenses and vehicle documents.
- Replace paper-based tickets with electronic citation systems.
- Improve transparency by digitally recording traffic stops and enforcement activities.
- Automate tracking of unpaid fines and repeat offenders.
- Provide traffic statistics and reports to support planning and decision-making.
- Allow citizens to check violations and pay fines online.

Significance of the Project

The RTCS project is important for improving road safety and institutional efficiency in the Amhara Region. It will reduce document forgery, improve transparency, and strengthen accountability in traffic enforcement. By digitizing traffic management, officers can work faster and more accurately, reducing congestion and improving public service delivery.

The system will also generate reliable traffic data that helps policymakers design better road safety strategies. In the long term, RTCS will support Ethiopia's digital transformation and contribute to safer, more organized, and more efficient road transportation.

2: Afalgugn – Lost and Found Family and Property System

Introduction

In Ethiopia, many families face emotional and practical challenges when loved ones or important belongings go missing. Current search methods rely on word-of-mouth and manual police reports, which are slow and limited. Afalgugn (“to find again”) is a mobile application designed to connect people who lost family members or property with those who may have found them through a centralized digital platform.

Statement of the Problem

There is no unified digital system for reporting and matching missing persons or lost property. Information is scattered, communication is slow, and many cases remain unresolved. Families experience stress, and institutions rely on paper-based processes.

Objectives

General Objective

To develop a mobile-based system that helps communities report, track, and match missing persons and lost property efficiently.

Specific Objectives:

- Enable digital reporting with photos and location details.

- Provide matching and notification features.
- Support collaboration with police and hospitals.
- Ensure secure and verified information sharing.

Significance of the Project

Afalgugn improves coordination, reduces search time, and strengthens community cooperation. It promotes compassion, trust, and faster reunification of families while supporting institutions with organized digital records.

3: EthioStay – Nearby Room, Hotel and Guest House Finder

Introduction

Finding temporary accommodation in Ethiopia is often difficult and time-consuming. Travelers, students, and workers depend on informal methods to locate rooms. EthioStay is a GPS-based mobile application that helps users quickly find nearby hotels, guest houses, and rooms with clear price and availability details.

Statement of the Problem

There is no reliable, centralized digital platform for searching and comparing accommodations in Ethiopia. This leads to wasted time, higher costs, and limited visibility for property owners.

Objectives

General Objective:

To develop a location-based mobile app that connects users with nearby accommodation options quickly and reliably.

Specific Objectives:

- Provide verified listings with price and availability.
- Allow property owners to manage digital listings.
- Enable easy search and comparison features.

Significance of the Project

EthioStay saves time, improves transparency, and supports local tourism and small businesses. It creates a convenient and organized way to access temporary housing in Ethiopia.